Continental
The Future in Motion

Automotive Software Architectures

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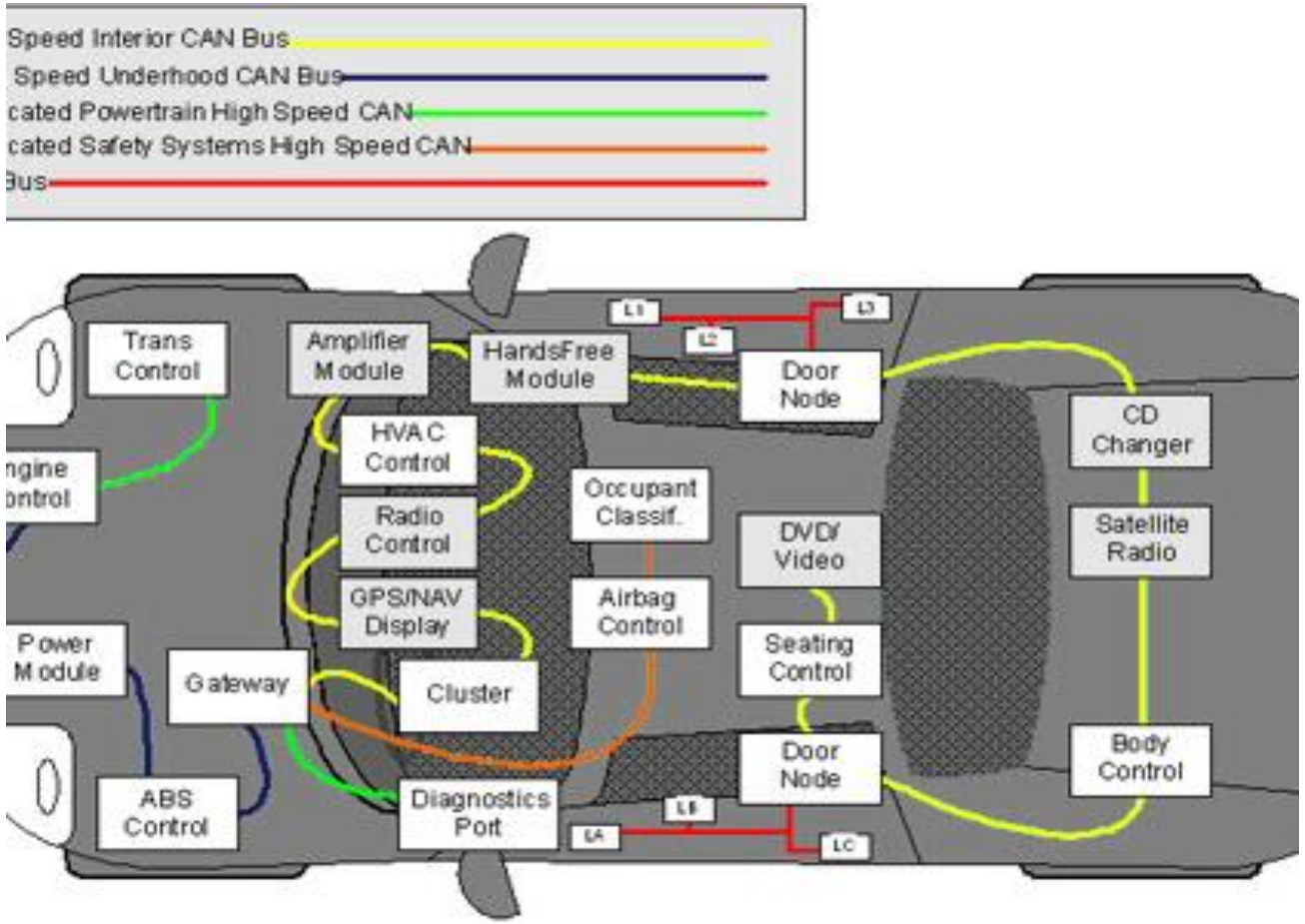
Agenda

1. Introduction and definitions
2. Architecture
3. Contemporary Software Architectures:
   I. Federated
   II. Centralized

1. Introduction and definitions

- Software architecting starts with the very first requirement and ends with the last defect fix in the product, although its most intensive period is in the early design stage where the architects decide upon the high-level principles of the system design

- Software architecture refers to the high-level structures of a software system, the discipline of creating such structures, and the documentation of these structures. These structures are needed to reason about the software system



2. Architecture

1. **Software components/Blocks** — pieces of software packaged into subsystems and components based on their logical structure. Examples of such components could be UML/C++ classes, C code modules, and XML configuration files.

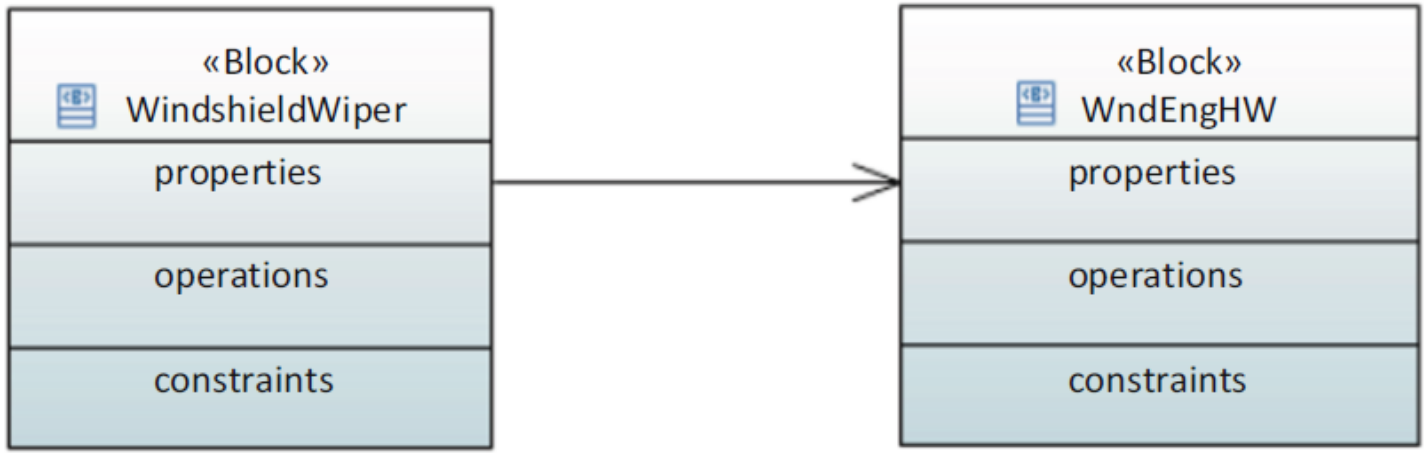
2. **Hardware components/Electronic Control Units** — elements of design of the computer system (or platform) on which the software is executed. Examples of such elements include ECUs, communication buses, sensors and actuators.

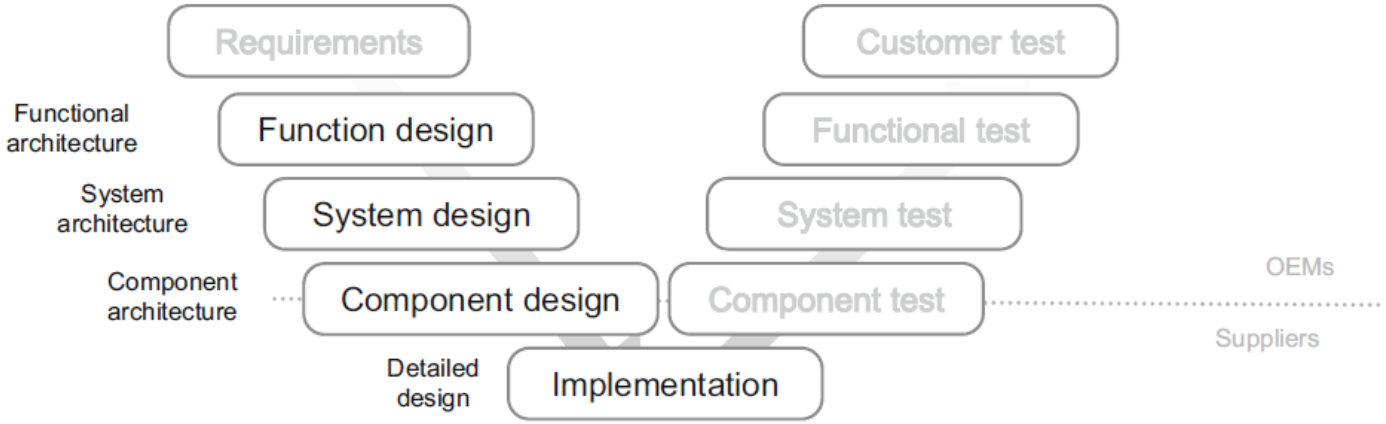
3. **Functions** — elements of the logical design of the software described in terms of functionality, which is then distributed over the software components/blocks. Examples of such elements are software functions, properties and requirements.

An architectural view is *a representation of one or more structural aspects of an architecture that illustrates how the architecture addresses one or more concerns held by one or more of its stakeholders*

Internal

3. Architectural Principles and development process



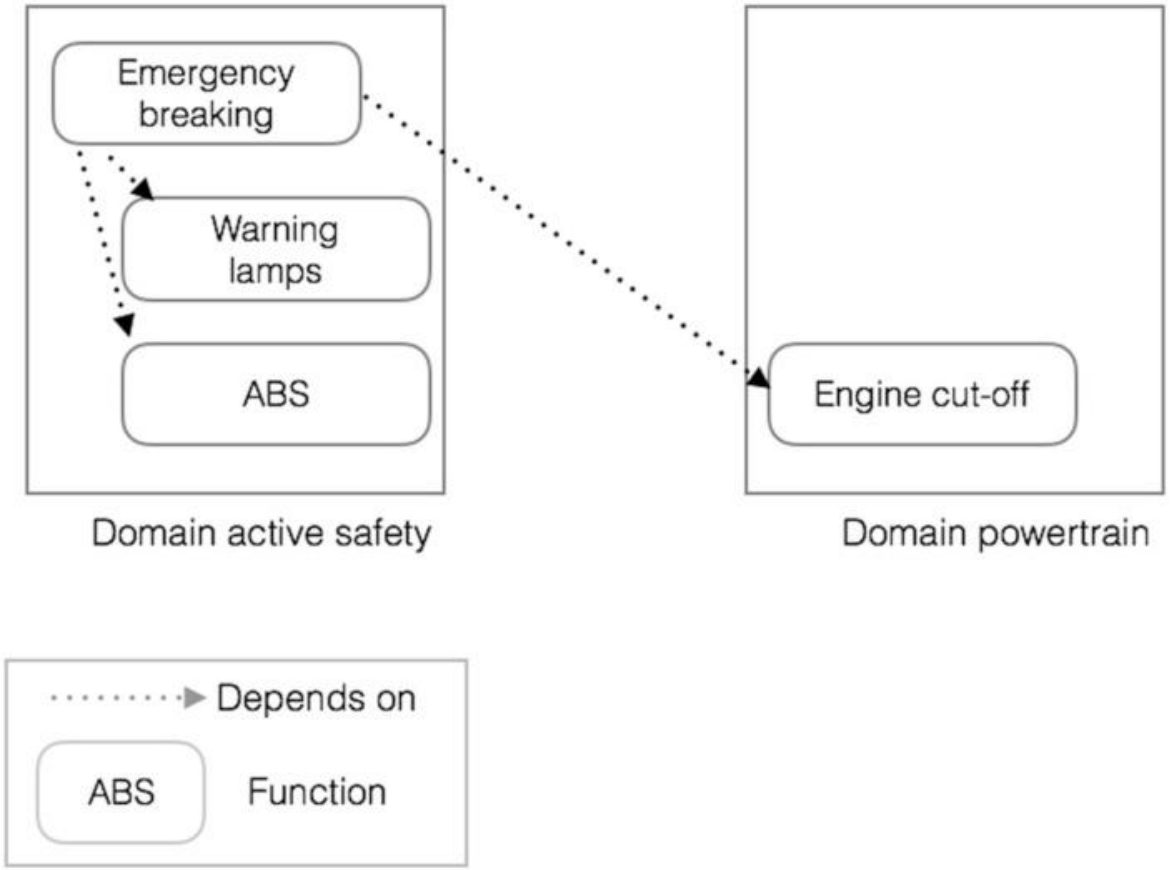


Architectural Views

› ***Functional View***

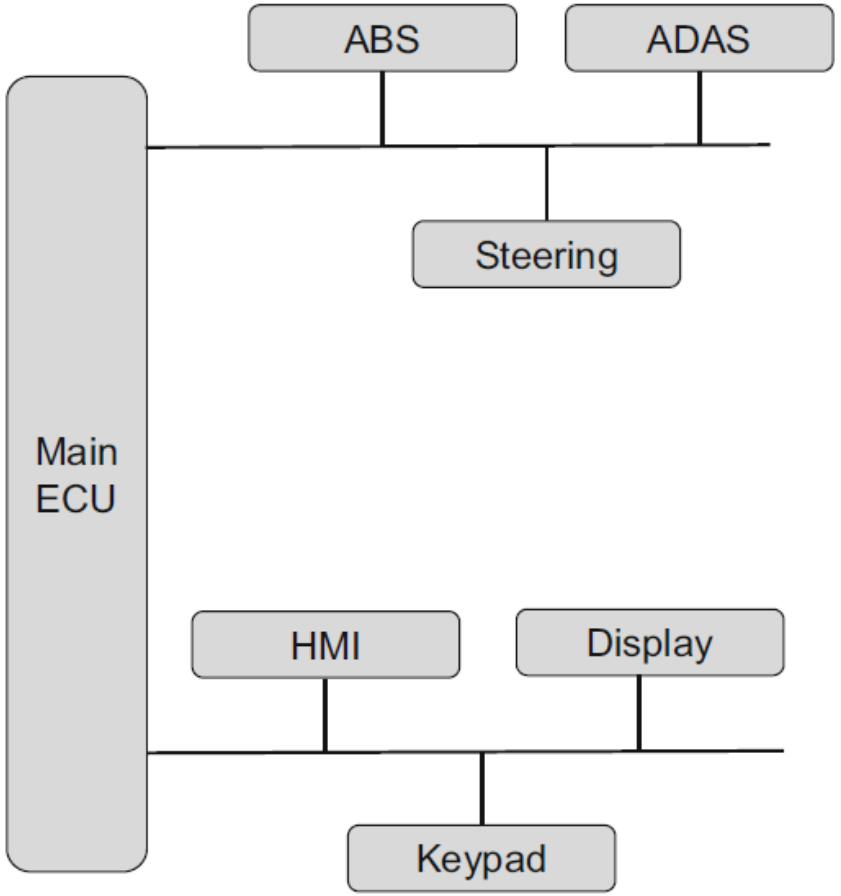
› **The functional view, often abbreviated to *functional architecture*, is the view where the focus is on the functions of the vehicle and their dependencies on one another**

1. Powertrain — grouping the elements related to the powertrain of the car—engine, engine ECU, gearbox and exhaust.
2. Active Safety — grouping the elements related to safety of the car — ADAS
3. (Advanced Driver Assistance Systems), ABS (Anti-lock Braking System) and similar.
4. Chassi and body — grouping the elements related to the interior of the carseats, windows and other (which also contain electronics and software actuators/sensors).
5. Electronic systems — grouping the elements related to the functioning of the car's electronic system—main ECU, communication buses and related.

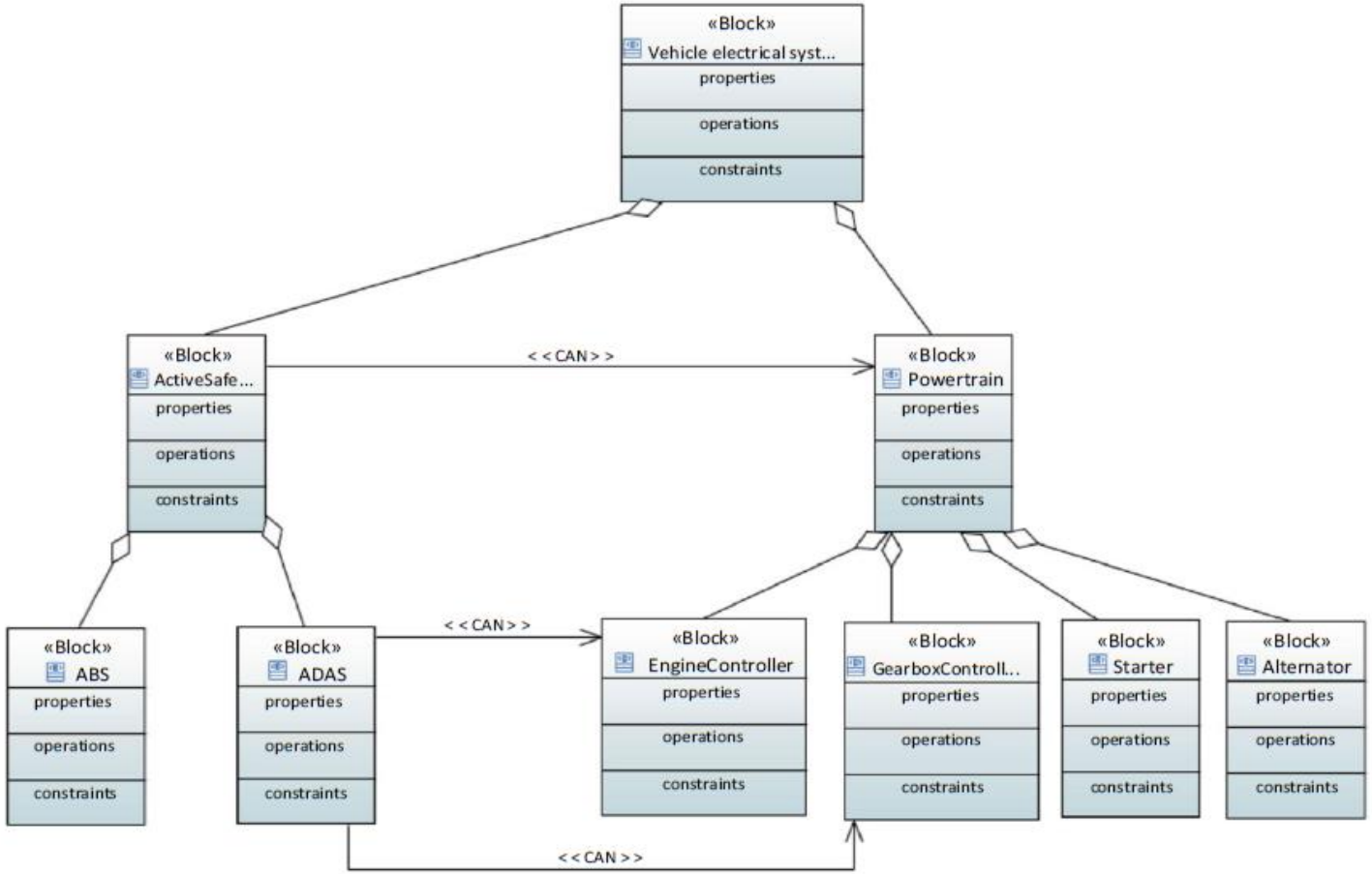


Architectural Views

Physical System View

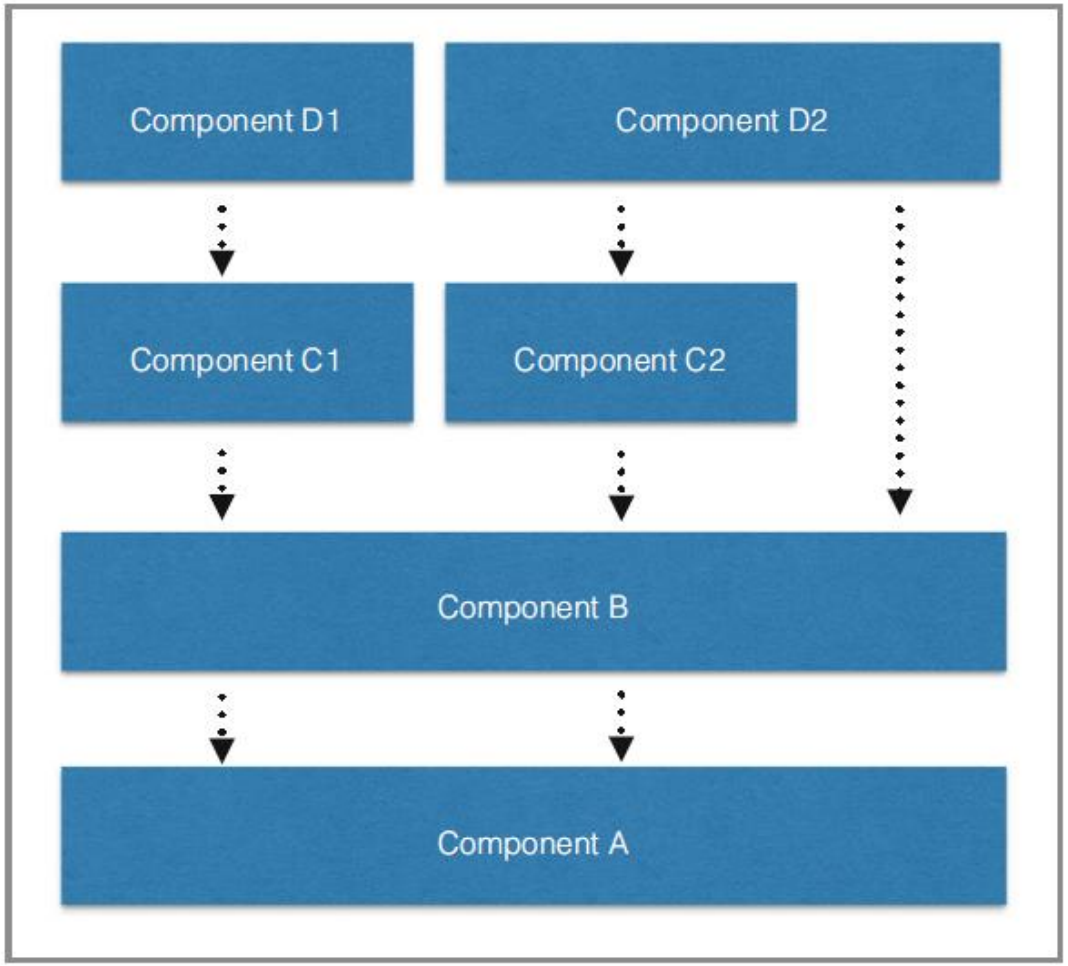


Logical view

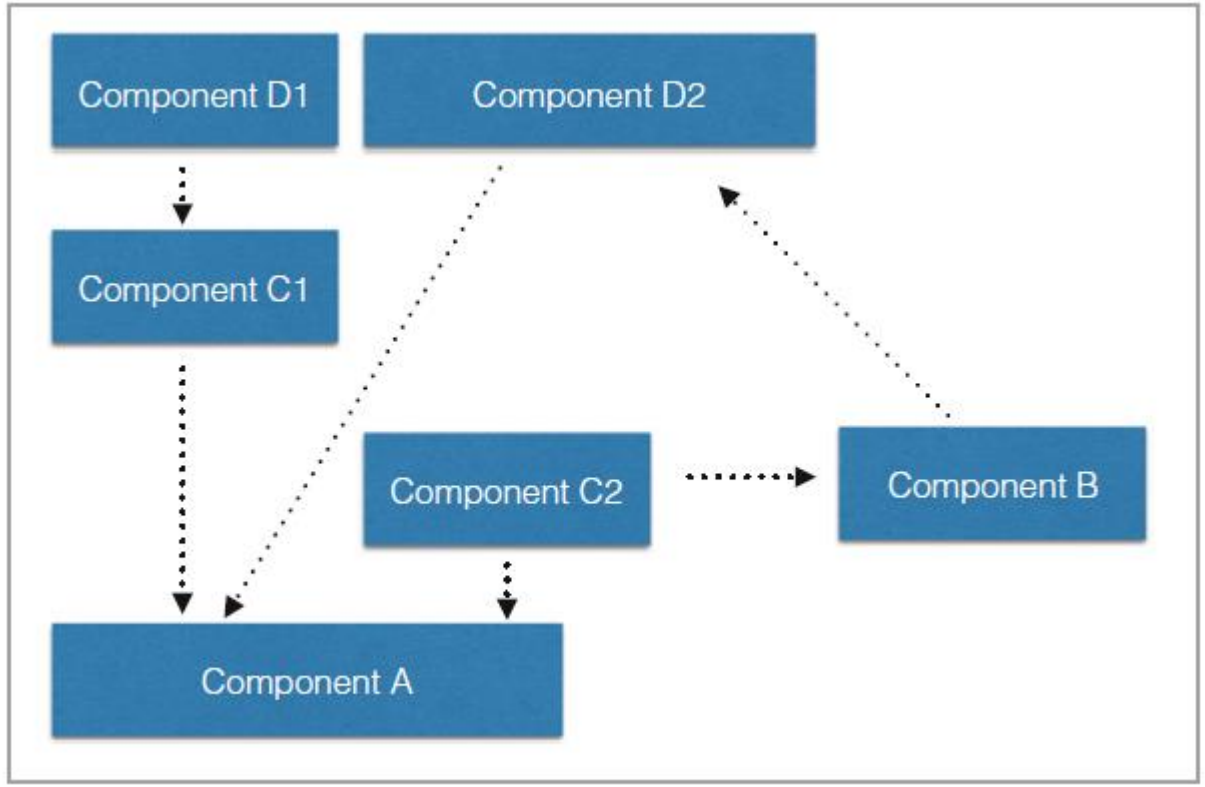


Architectural styles

Layered Architecture

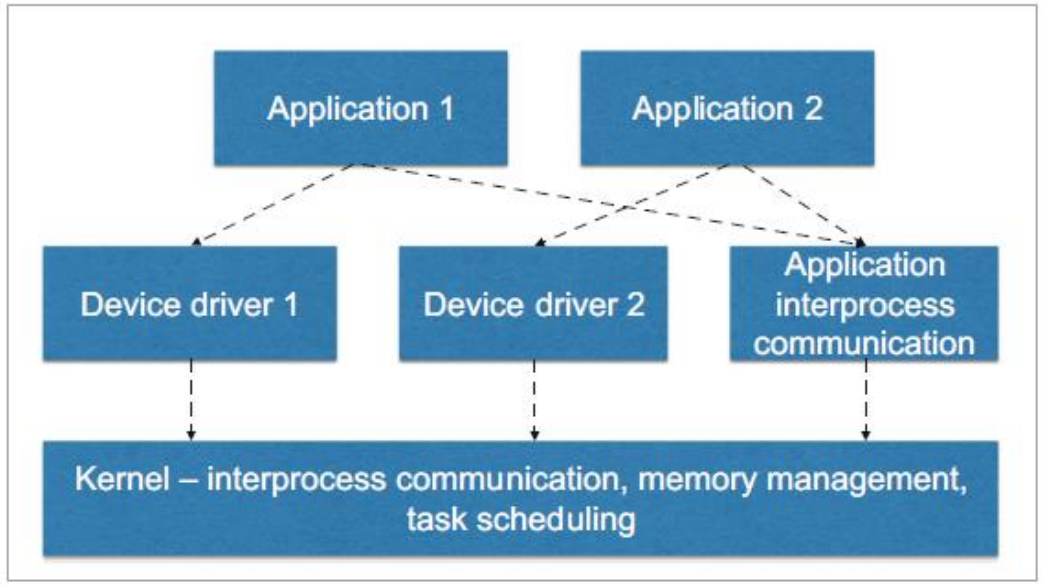


Component Based

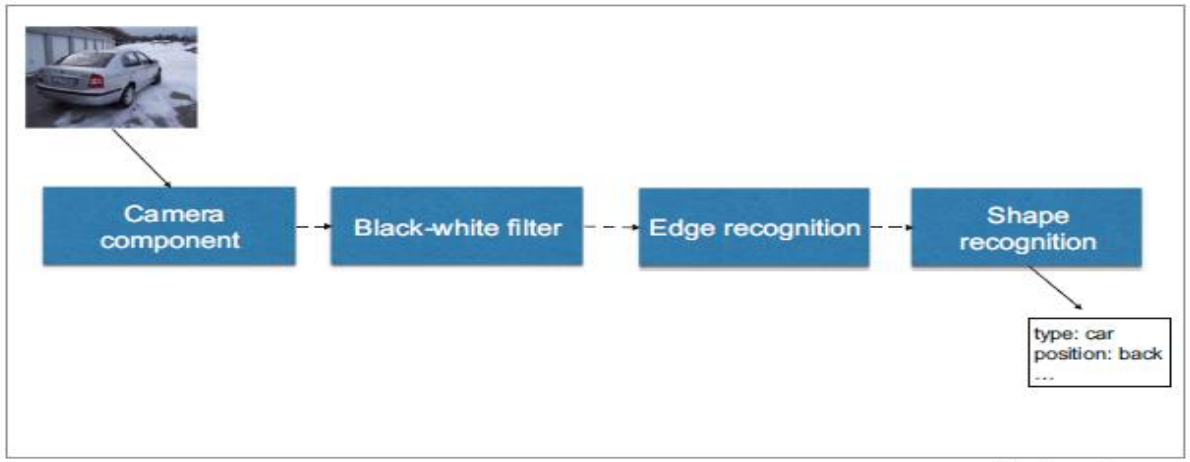


Architectural styles

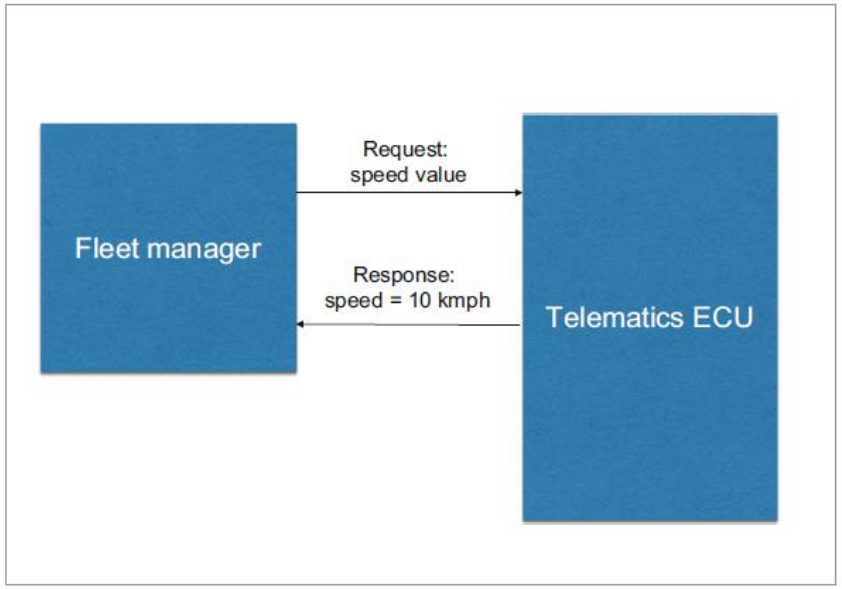
Mikrokernel



Pipes and Filters

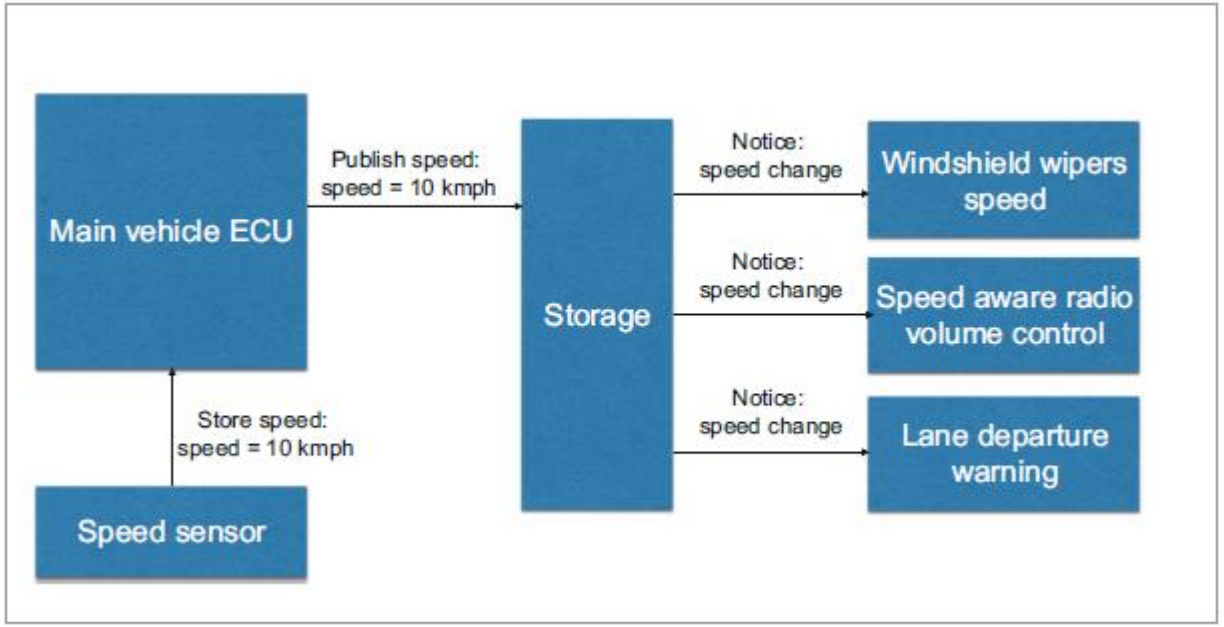


Client–Server

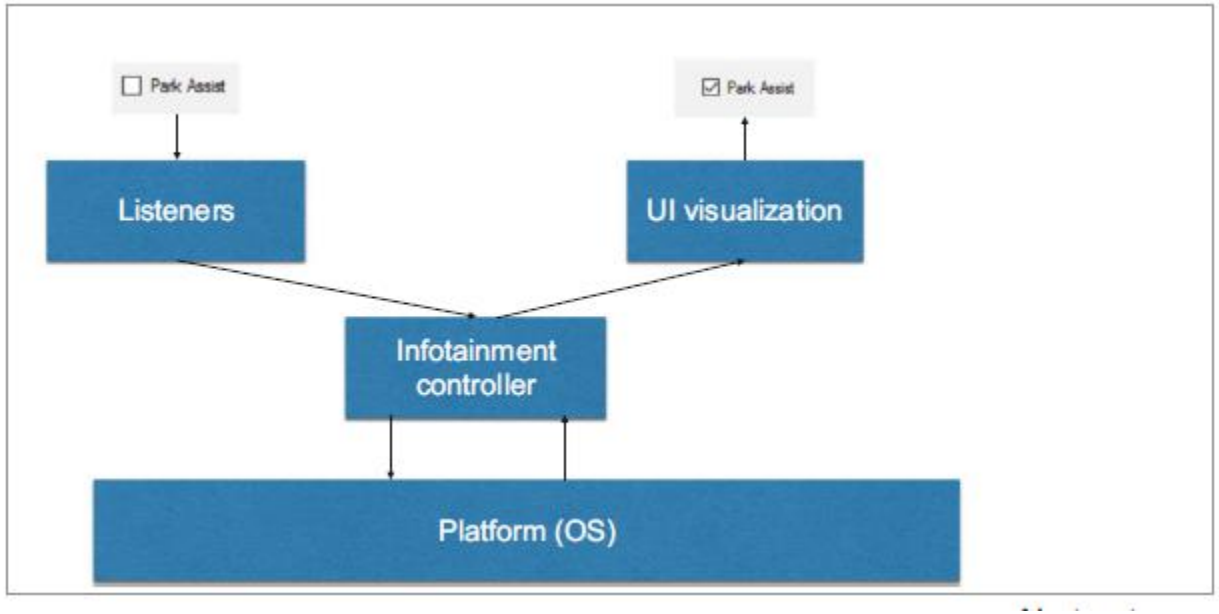


Architectural styles

Publisher–Subscriber

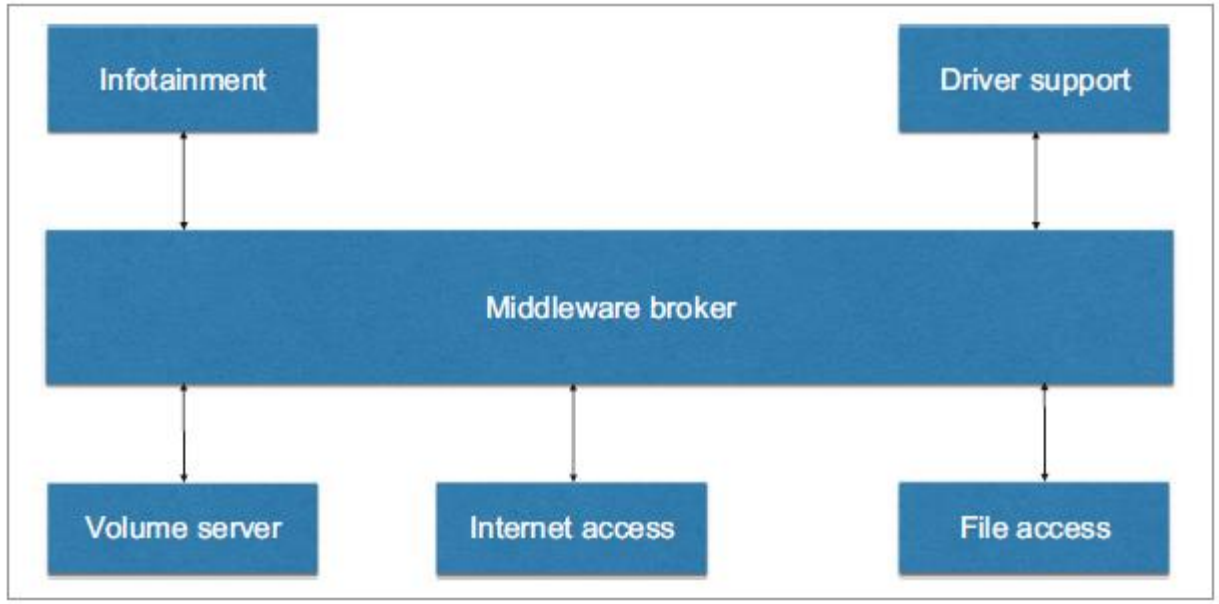


Event–Driven

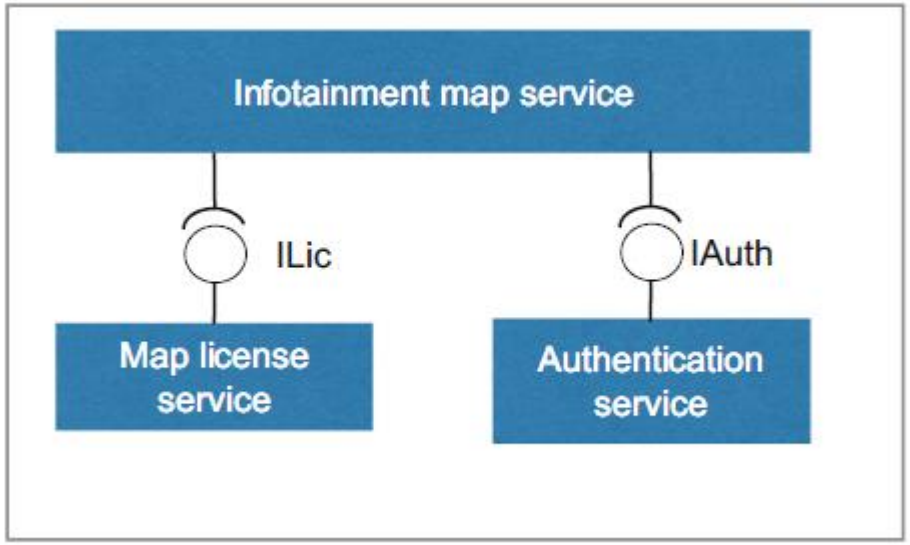


Architectural styles

Middleware



Service-Oriented

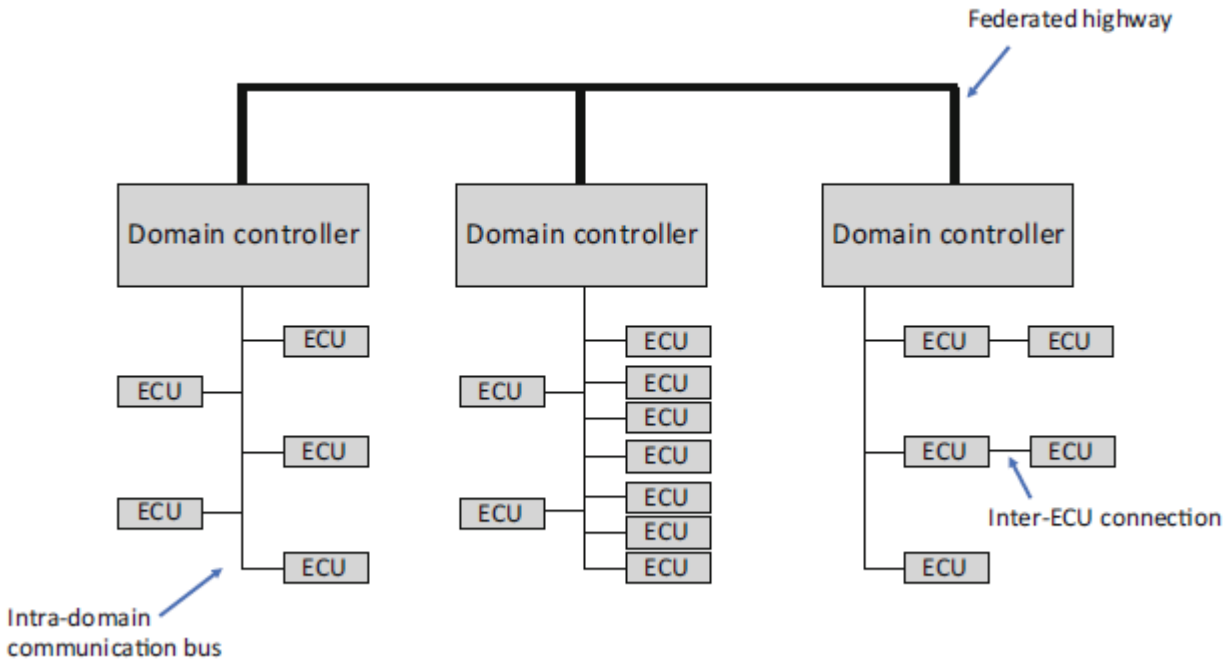


Contemporary Software Architectures

Automotive software architectures evolve together with the evolution of the electronics in vehicles:

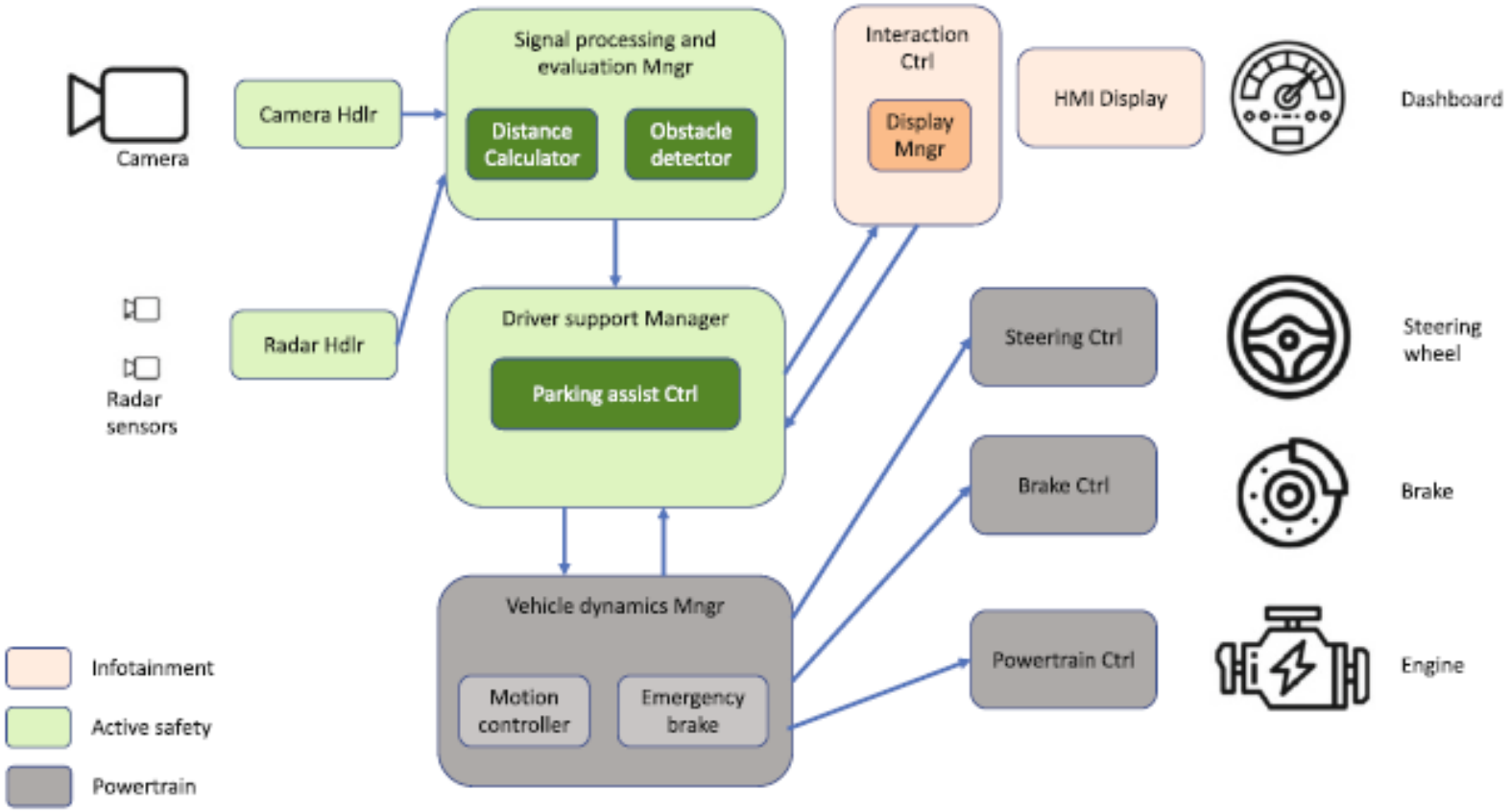
Federated software architecture:

- The concept of federated software comes essentially from the field of enterprise computing, where the enterprise system architecture reflects the organization's structure. In the automotive field, the federation reflects the domains of the vehicle's system rather than the organization. The most common domains are:
    - Active safety – responsible for such functions as collision avoidance or emergency braking
    - Infotainment – responsible for displays or connections to mobile phones
    - Powertrain – responsible for the engine and gearbox's software

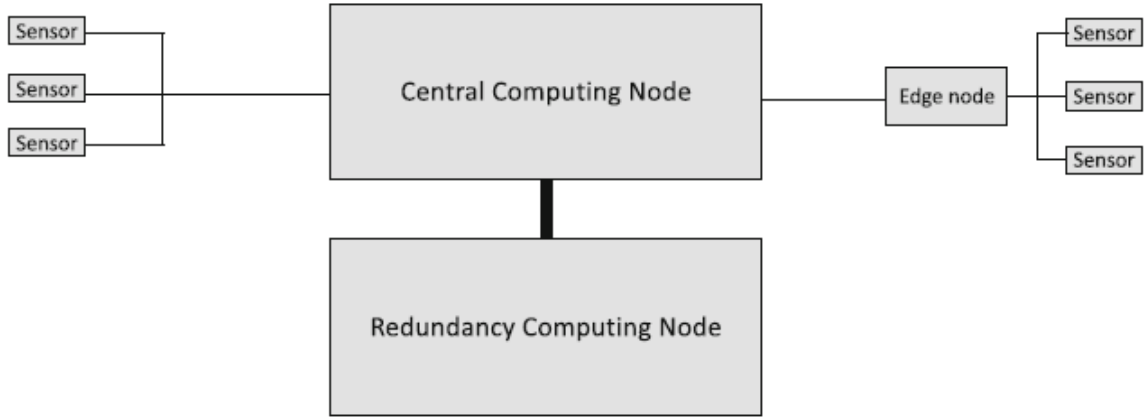


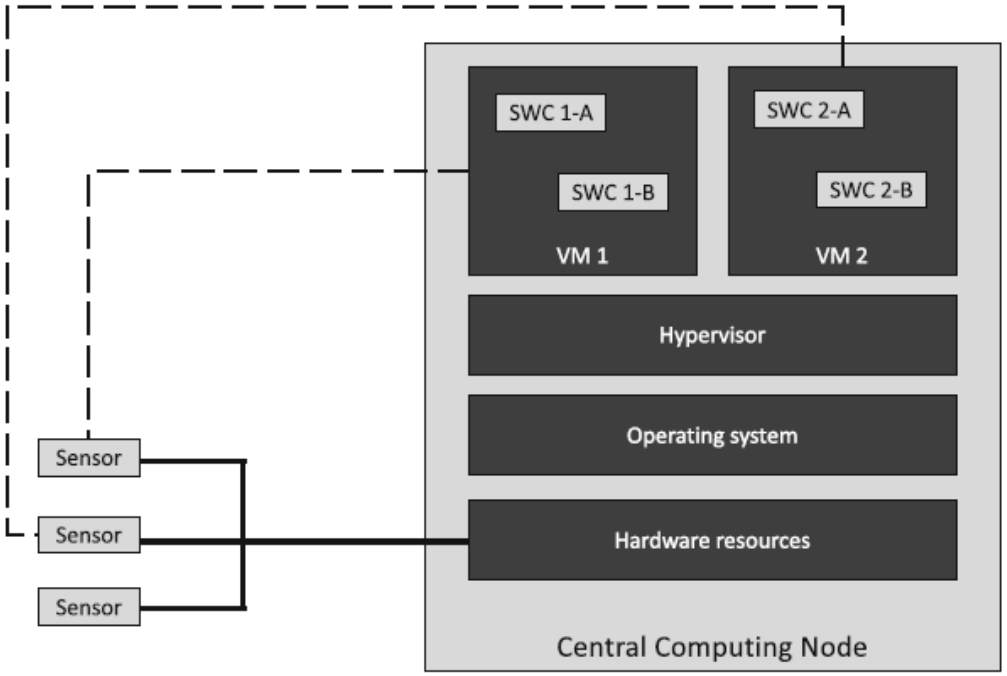
Centralized software architecture

Contemporary Software Architectures



Centralized Software Architectures





Thank you!